Missionaries and Cannibals Problem

AIM:

To write a Python program to solve the Missionaries and Cannibals Problem while ensuring that the missionaries are never outnumbered by cannibals on either bank when missionaries are present.

Algorithm:

1. Start the program.
2. Initialize the numbers of missionaries and cannibals on both riverbanks.
3. Place the boat on the starting bank.
4. Generate possible boat movements carrying one or two people.
5. Check that no bank has more cannibals than missionaries when missionaries are present.
6. Reject invalid or previously visited states.
7. Continue until all missionaries and cannibals reach the destination bank safely.
8. Display the sequence of valid states.
9. Stop the program.

Code :

exp05_missionaries_cannibals.py > is_valid

```python
from collections import deque

def is_valid(m, c):
    if m < 0 or m > 3 or c < 0 or c > 3:
        return False
    if m > 0 and m < c:
        return False
    m_right = 3 - m
    c_right = 3 - c
    if m_right > 0 and m_right < c_right:
        return False
    return True

def get_successors(state):
    m, c, boat = state
    moves = [(1, 0), (2, 0), (0, 1), (0, 2), (1, 1)]
    successors = []

    for dm, dc in moves:
        if boat == 1:
            nm, nc = m - dm, c - dc
            n_boat = 0
            action = f"Cross {dm}M and {dc}C to Right"
        else:
            nm, nc = m + dm, c + dc
            n_boat = 1
            action = f"Cross {dm}M and {dc}C to Left"

        if is_valid(nm, nc):
            successors.append(((nm, nc, n_boat), action))

    return successors

def solve_missionaries_cannibals():
    start_state = (3, 3, 1)
    goal_state = (0, 0, 0)
    queue = deque([(start_state, [])])
```

Output :

```
                        > & C:\python311\python.exe d:/CSA1702-AI/Experiments/exp05_missionaries_cannibals.py
issionaries and Cannibals Solution Steps:

tep 0: Left Bank: (3M, 3C), Right Bank: (0M, 0C), Boat: Left -> Cross 0M and 2C to Right
tep 1: Left Bank: (3M, 1C), Right Bank: (0M, 2C), Boat: Right -> Cross 0M and 1C to Left
tep 2: Left Bank: (3M, 2C), Right Bank: (0M, 1C), Boat: Left -> Cross 0M and 2C to Right
tep 3: Left Bank: (3M, 0C), Right Bank: (0M, 3C), Boat: Right -> Cross 0M and 1C to Left
tep 4: Left Bank: (3M, 1C), Right Bank: (0M, 2C), Boat: Left -> Cross 2M and 0C to Right
tep 5: Left Bank: (1M, 1C), Right Bank: (2M, 2C), Boat: Right -> Cross 1M and 1C to Left
tep 6: Left Bank: (2M, 2C), Right Bank: (1M, 1C), Boat: Left -> Cross 2M and 0C to Right
tep 7: Left Bank: (0M, 2C), Right Bank: (3M, 1C), Boat: Right -> Cross 0M and 1C to Left
tep 8: Left Bank: (0M, 3C), Right Bank: (3M, 0C), Boat: Left -> Cross 0M and 2C to Right
tep 9: Left Bank: (0M, 1C), Right Bank: (3M, 2C), Boat: Right -> Cross 1M and 0C to Left
tep 10: Left Bank: (1M, 1C), Right Bank: (2M, 2C), Boat: Left -> Cross 1M and 1C to Right
tep 11: Left Bank: (0M, 0C), Right Bank: (3M, 3C), Boat: Right -> Goal Reached
S D:\CSA1702-AI\Experiments>
```

RESULT: Thus, the Python program to solve the Missionaries and Cannibals Problem was executed successfully.